Paper II Audit -- P1: max Phi_tau over chordal graphs vs M(n)

Test	P1 -- max Phi_tau over chordal graphs vs M(n)
Gates	F, G, and the main kill-switch (Theorem 1.2)
Evidence	EXHAUSTIVE_FINITE_VERIFICATION (n<=6) + complete-split scan + random chordal
tau3* method	auditor's own cover-LP via scipy HiGHS (independent)
Master seed	20260710
Environment	Python 3.14.4 / networkx 3.6.1

1. Claim under test (Theorem 1.2)

max over chordal G on n vertices of Phi_tau(G)=|E|-2*tau3*(G) equals M(n)=floor((2n+1)^2/24), attained by a complete-split. tau3* is recomputed from the auditor's own cover-LP (scipy HiGHS), independent of Paper II code.

2. Kill-switch (declared before running)

H1a no chordal G exceeds M(n); **H1b** exhaustive max = M(n) for n<=6; **H1c** the maximiser is a complete-split. Falsify on any hit.

3. Result

Violations of the upper bound: **0**. Exhaustive max = M(n) for all n<=6: **True**. Maximiser is a complete-split (n<=6): **True**. Complete-split scan matches M(n) for all n<=60: **True**.

Verdict: VERIFIED (n<=6 exhaustive) + SUPPORTED (larger n): no chordal graph exceeds M(n); exhaustive max = M(n) and is attained by a complete-split; complete-split scan matches M(n) for n<=60.

4. Exhaustive enumeration (n<=6, all labelled chordal graphs)

n	#chordal	max Phi_tau	M(n)	max=M(n)?	maximiser=complete-split?
1	1	0.0	0	yes	yes
2	2	1.0	1	yes	yes
3	8	2.0	2	yes	yes
4	61	3.0	3	yes	yes
5	822	5.0	5	yes	yes
6	18154	7.0	7	yes	yes

5. Scope

For n<=6 the enumeration is EXHAUSTIVE over all labelled graphs, so the equality max=M(n) and the complete-split attainment are CONCLUSIVE for those n. The complete-split closed-form scan (n<=60) and random chordal graphs (n=7..16) are supporting evidence; a bounded search finding no counterexample is not a proof of the universal statement. No counterexample was found.